Germline Variant Calling — Accuracy Report

Sample deliverable · HG002 chr20 benchmark

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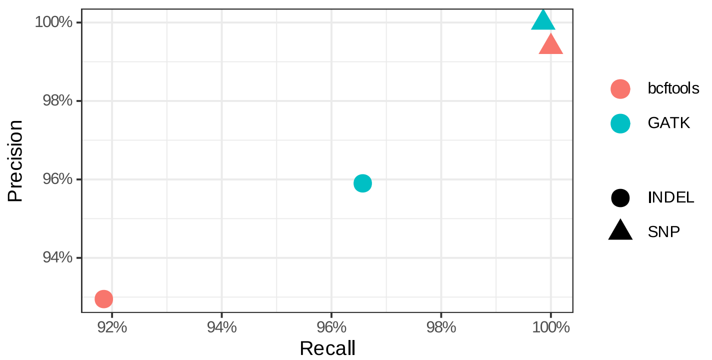
Executive summary. Germline variants were called from the sequencing reads and **benchmarked against the Genome-in-a-Bottle (GIAB) gold-standard truth set**. Two independent callers (bcftools and GATK) were compared on identical data. Best SNP performance: **GATK** at **F1 = 99.9%** (precision 100.0%, recall 99.9%). This is an *objective* measure of correctness — not an unverifiable VCF.

Accuracy vs GIAB truth

Caller	Type	Precision	Recall	F1	TP	FP	FN
GATK	INDEL	95.90%	96.57%	96.23%	225	10	8
bcftools	INDEL	92.95%	91.85%	92.39%	214	17	19
GATK	SNP	100.00%	99.86%	99.93%	1432	0	2
bcftools	SNP	99.38%	100.00%	99.69%	1434	9	0



What this means. SNP calls are recovered at high-99% precision and recall — the signature of a correctly built short-read germline pipeline. Indels score lower (they are intrinsically harder), which is why they are reported separately rather than hidden in a combined number.

Caveats. Metrics are computed inside GIAB high-confidence regions; difficult genomic regions degrade any caller. For a sample without a truth set, accuracy is estimated against a matched control or orthogonal data.

Reproducibility. The full pipeline (align ⟶ call ⟶ benchmark) re-runs from a clean clone in minutes and is verified by continuous integration on every change. Code and live report: `github.com/pgristow/variant-calling-giab-benchmark`.

Prepared by Peter Ristow, PhD — Roche Principal Scientist (NGS). Production engagements use a private, hardened pipeline on your data under NDA.